

LeetCode 931: Minimum Falling Path Sum

DP State Definition

The 2D array `dp` tracks the state. As noted:
"min cost to be at a cell"

Base Case Setup

The top row is copied over directly from the matrix:
"initialization cause top row khud hi minimum he"

```
class Solution {
public:
    int minFallingPathSum(vector<vector<int>>& matrix) {

        int n=matrix.size();
        int m=matrix[0].size();

        vector<vector<int>> dp(n,vector<int>(m,INT_MAX));
        //min cost to be at a cell

        for(int j=0;j<m;j++){
            dp[0][j]=matrix[0][j];
        }

        //initialization cause top row khud hi minimum he

        for(int r=0;r<n-1;r++){
            for(int c=0;c<m;c++){

                int nr=r+1;
                int nc1=c+1;
                int nc2=c-1;
                int nc3=c;

                if(nc1<m){
                    dp[nr][nc1]=min(dp[nr][nc1],dp[r][c]+matrix[nr][nc1]);
                }

                if(nc2>=0){
                    dp[nr][nc2]=min(dp[nr][nc2],dp[r][c]+matrix[nr][nc2]);
                }

                dp[nr][nc3]=min(dp[nr][nc3],dp[r][c]+matrix[nr][nc3]);

            }
        }

        int ans=INT_MAX;
        for(int i=0;i<m;i++){
            ans=min(ans,dp[n-1][i]);
        }
        return ans;
    }
};
```